



Optimal transport based pyramid graph kernel for autism spectrum disorder diagnosis

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ABSTRACT

Brain network, which characterizes the functional and structural interactions of brain regions with graph theory, has been widely utilized to diagnose brain diseases, such as autism spectrum disorder (ASD). It is a challenge to measure the network (or graph) similarity in brain network analysis. Graph kernel (i.e., kernel defined on graphs) offers an efficient tool for measuring the similarity of paired brain networks and yields the excellent classification performance in brain disease diagnosis. However, most of the existing graph kernels neglected the hierarchical architecture information of brain networks. To address this problem, in this paper, we propose an optimal transport based pyramid graph kernel for measuring brain network similarity and then apply it to brain disease classification. The main idea is to transform brain networks into pyramid structures, which reflect the hierarchical architecture information of the brain network with multi-resolution histograms. The optimal transport distance in pyramid structures is calculated for measuring transport costs between paired brain networks. Finally, the optimal transport based pyramid graph kernel is computed based on this optimal transport distance. To evaluate the effectiveness of the proposed optimal transport based pyramid graph kernel, the extensive experiments are performed in functional magnetic resonance imaging data of brain disease from the Autism Brain Imaging Data Exchange database. The experimental results show that our proposed optimal transport based pyramid graph kernel outperforms the state-of-the-art methods in ASD classification tasks.

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1. Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disease which is defined by repetitive behaviour and impairments in social communication, and by extremely restricted interests and sensory behaviour beginning early in life. The centers for disease control and prevention reports that there are about 1.5% of American children suffering from ASD, which brings severe financial burdens to society and the family [1]. Children with ASD usually have problems with emotional, social, and communication skills. They may repeat certain behaviors and do not want to change in their daily activities. Many children with ASD also have different ways of paying attention, learning, or reacting to things. ASD usually begins during early childhood and typically lasts throughout a person's whole life. About 2/3 of the children with ASD cannot live independently after adulthood and need lifelong care and maintenance. The related studies showed that early intervention treatment can improve the development of children with ASD [2]. Hence, it is

crucial for possible improvement and early treatment of ASD progression to accurately diagnose ASD.

Functional magnetic resonance imaging (fMRI) as one of the non-invasive imaging technologies is widely used in the studies related to brain diseases (e.g., ASD) where fMRI quantifies neural activity by mapping functional connectivity patterns of the brain. These functional connectivity patterns can be described as a brain network with graph theory [3], which is helpful for investigating the neuropathological mechanism of brain disease via detecting the connectivities in brain network. For example, studies based on brain functional network showed that ASD patients had significantly different brain functional connectivities in hippocampus when compared with normal controls [4].

Brain network characterizes the functional and structural interactions of brain regions by using graph theory, where brain regions correspond to nodes and functional or structural connectivities of the paired brain regions are treated as edges. Brain network is widely utilized to diagnose brain diseases, such as ASD [5] and Alzheimer's disease [6]. In these diagnosis tasks, network characteristics, e.g., functional connectivity [4] and ordinal pattern [7], are calculated in brain network. Then, these network characteris-

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tics as feature vectors are applied to machine learning classifiers to classify brain diseases. Regarding brain network as feature vectors neglects the topological information of brain network. However, it is a challenging task for feature vector based methods to measure the topological similarity of the paired networks (or graphs). To address this problem, numerous methods are presented to measure network similarity.

Among various methods, the kernel methods, especially graph kernel, provides a promising tool to measure the similarity of paired networks. Graph kernel measuring network topological similarity has been widely applied to brain network analysis and yields excellent classification performance in brain disease diagnosis. However, most of the existing graph kernels are constructed on single layer graph and ignore the information of graph hierarchical architecture. To utilize the information of graph hierarchical architecture, many representational methods are proposed for pattern recognition tasks on graphs. For example, Liu et al. [8] proposed a whole brain hierarchical network and combined it with multiple kernel learning where Gaussian kernels are used to measure the similarity between paired networks. Then, they utilized the kernels to diagnose Alzheimer's Disease. In whole brain hierarchical network, the hierarchical structure was constructed by comparing the digits of brain regions and ignored the structural information in network. Nikolentzos et al. [9] transformed graphs into multi-resolution histograms which were leveraged successfully in image processing field and computer vision. They used pyramid structure based on multi-resolution histograms to acquire graph hierarchical representation. Further, they utilized the intersections of multi-resolution histograms to design graph kernel and applied this graph kernel to the classification task of bioinformatics datasets, obtaining promising classification performance. However, their proposed graph kernel focused on the local information of multi-resolution histograms (i.e., histogram intersections), and hence ignored the global information of multi-resolution histograms (i.e., all histograms).

Optimal transport (OT) distance, which is also called Wasserstein distance or Earth Mover's distance, expresses excellent performance in comparing two histograms [10]. OT distance draws great attention in graph data. For example, Vayer et al. [11] proposed fused Gromov-Wasserstein for computing the distances between graph objects and obtained the superior classification results. Xu et al. [12] proposed a novel Gromov-Wasserstein learning framework to jointly match graphs and learn embedding vectors. They used Gromov-Wasserstein discrepancy to measure the similarity between graphs and find the correspondence of graphs using the learned optimal transport. In these studies, OT distance was indirectly used to measure the similarity between the paired graphs. OT distance can also be used to design graph kernel for directly measuring graph similarity. Togninalli et al. [13] proposed Wasserstein Weisfeiler-Lehman graph kernel by combining Weisfeiler-Lehman scheme with OT distance. However, in graph measurement (i.e., graph kernel) based on OT distance, most of the existing studies ignore the hierarchical architecture information of the graph.

To address these problems, we develop an optimal transport based pyramid graph kernel for measuring brain network similarity. In this work, we construct brain networks for all subjects based on fMRI data, and then represent each brain network as a pyramid structure which reflects the hierarchical architecture information of the network with multi-resolution histograms. In Fig. 1, we plot the pyramid structure and multi-resolution histograms for one ASD patient and one normal control (NC). Each layer of the pyramid structure is multi-resolution histograms. From Fig. 1, we can find that ASD patient is different from NC in the pyramid structure. Then, we calculate the OT distance in pyramid structures between paired subjects (e.g., ASD patient or NC). Finally, we use these OT distances to calculate optimal transport based pyramid graph ker-

nel, and further develop a brain disease diagnosis framework based on the proposed kernel using resting-state fMRI (rs-fMRI) data.

Our work includes three major contributions.

- A new network descriptor (i.e., pyramid structure) is exploited to express brain network. Pyramid structure can depict the hierarchical architecture of brain network.
- We propose an optimal transport based pyramid graph kernel and use this graph kernel to develop a brain disease diagnosis framework based on rs-fMRI.
- We evaluate the proposed graph kernel on five ASD datasets from Autism Brain Imaging Data Exchange (ABIDE) database, with the experimental results demonstrating the performance of our proposed method.

The rest of the paper is organized as follows. In Section 2, we introduce the related works. In Section 4, we introduce the materials used in our work and present the proposed method. In Section 5, we introduce the experimental settings and results. In Section 6, we present the discussion for the experimental results. At last, we conclude this paper in Section 7.

2. Related works

2.1. Graph kernel

Graph theory is used to characterize the functional and structural interactions of brain regions in brain network, which is utilized to investigate the pathological mechanism related to brain diseases (e.g., ASD). For example, Delbruck et al. [14] investigated atypical pathways in brain networks and found that children with ASD had decreased functional connectivity in lateral occipital cortex and fusiform gyrus compared to age-matched controls. Cerliani et al. [15] found the increased functional connectivities of subcortical and cortical resting-state networks in ASD. However, these studies focused on network characteristics (e.g., functional connectivity and clustering coefficient) of the brain network and ignored the topological structure information in brain network.

Graph kernel, which offers a promising tool for investigating topological structure information of brain network, measures the topological similarity of the paired brain networks. Graph kernel has produced excellent performance on graph classification tasks [16,17]. Brain network is a kind of graph data. Hence, graph kernel can be applied to brain network to diagnose brain diseases. A variety of graph kernels have been proposed in different network topological characteristics which involve paths, walks, trees and subgraphs. For example, shortest-path kernel [18] measured the similarity between the paired graphs using graph shortest path characteristic. Random walk graph kernel [19], quantum-inspired graph kernel [20], and fast quantum walk kernel [21] measured graph similarity on walk characteristics. Random walk graph kernel calculated the matching random walks between a pair of graphs. Quantum-inspired graph kernel extracted a commute time minimum spanning tree from the complete weighted graph and the spanning tree is probed by a discrete-time quantum walk. Fast quantum walk kernel measured the similarity of the neighborhood-pair substructure between two nodes by using the superposition amplitude of quantum walks where the quantum interference can provides more information on the substructures. Tree pattern kernels [22], Weisfeiler-Lehman graph kernel [23], and its variant [13] utilized the tree pattern characteristics to measure the similarity between graphs. Subgraph matching kernels [24] used subgraph characteristics to measure the similarity of graphs. Pyramid Match kernel [9] utilized histogram intersections as network characteristics to measure the similarity between paired graphs. However, these graph kernels ignored the hierarchi-

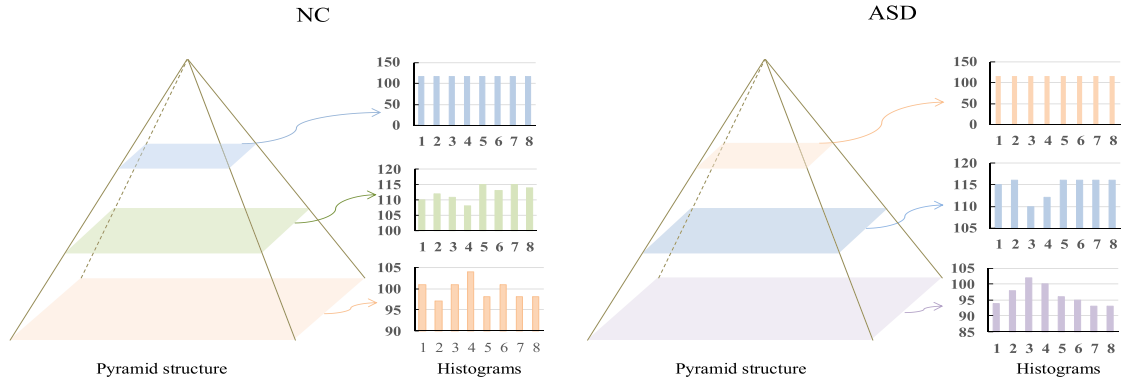


Fig. 1. Pyramid structure and multi-resolution histograms of ASD patient and normal control. ASD is autism spectrum disorder. NC is normal control.

cal architecture of brain network and global information on hierarchical architecture.

2.2. Optimal transport distance

OT was originally proposed to investigate the translocation of masses [10]. In mathematics, OT is utilized to measure the difference between two probability distributions or histograms after giving ground metric. The distance defined by OT is also called Wasserstein distance or Earth Mover's distance. Recently, OT draws great attention to image processing, computer vision and graph similarity measurement. For example, Isabelle et al. [25] used a multi-marginal OT approach to determine the shape of the interior interface. Zheng et al. [26] applied semi-dual OT to photo-realistic face super-resolution. Excepting image processing and computer vision, OT has inspired great attention in graph structure data.

Graph structure data can be represented as probability distributions with the related graph or network embedding methods. Hence, OT distance can be used to measure the similarity between graphs (i.e., define graph kernels) and capture graph geometric information. For example, Giannis et al. [9] utilized the eigen-decomposition of adjacency matrix to represent the graphs as bag-of-vectors and then used OT distance to measure the similarity of bag-of-vectors between two graphs. Togninalli et al. [13] used Weisfeiler-Lehman scheme to produce the node label sequences and then utilized the OT distance to measure the similarity of label sequences between two graphs. Vayer et al. [11] designed fused Gromov-Wasserstein distance to measure the similarity between structured data. However, these OT methods based on graph data neglected the hierarchical architecture information of the graph and the positive definition of these methods are hardly guaranteed.

To address these problems, in this paper we propose optimal transport based pyramid graph kernel for measuring the similarity between the paired brain networks.

3. Preliminary

3.1. Graph kernels

Graph kernels are a class of kernel functions measuring the similarities between a pair of graphs (or networks). There exists a function ϕ , which can implicitly map graph data set $\mathcal{G}$ into a Hilbert space $\mathcal{H}$, $\phi: \mathcal{G} \rightarrow \mathcal{H}$. In $\mathcal{G}$, the graph kernel $\mathcal{K}: \mathcal{G} \times \mathcal{G} \rightarrow \mathbb{R}$ is a function related to $\mathcal{H}$, given two graphs G_1 and G_2 , $G_1, G_2 \in \mathcal{G}$, graph kernel $\mathcal{K}$ can be interpreted as a dot product in high dimensional space $\mathcal{H}$, $\mathcal{K}(G_1, G_2) = \langle \phi(G_1), \phi(G_2) \rangle_{\mathcal{H}}$. The kernel function $\mathcal{K}$ is symmetric, i.e., $\mathcal{K}(G_1, G_2) = \mathcal{K}(G_2, G_1)$. If the Gram matrix $\mathbf{K} \in \mathbb{R}^{N \times N}$ defined by $\mathbf{K}(i, j) = \mathcal{K}(G_i, G_j)$ for any graph $\{G_i\}_{i=1}^N$ (N is the number of graphs) is positive definite, then $\mathcal{K}$ is positive definite.

3.2. Optimal transport distance

The optimal transport distances are very powerful in measuring two probability distributions or histograms, especially when the probability space has geometrical structures. For r and c are two probability distributions from the simplex Σ_n and Σ_m ($\Sigma_m := \{x \in \mathbb{R}_+^m : x^T \mathbf{1}_m = 1\}$), we regard $\Gamma(r, c)$ as the set of all transportation plans of r and c , Γ is a $n \times m$ matrices.

$$\Gamma(r, c) = \{P \in \mathbb{R}_+^{n \times m} | P\mathbf{1}_m = r, P^T \mathbf{1}_n = c\} \quad (1)$$

Given a $n \times m$ cost matrix M defined on the ground distance $d(x, y)$, $x \in r, y \in c$, the total cost of mapping r to c with a transportation matrix P is represented as $\langle P, M \rangle$. The optimal transport distance is defined as:

$$d_{OT}(r, c) = \min_{P \in \Gamma(r, c)} \langle P, M \rangle \quad (2)$$

The optimal transport distance in Eq. (2) is a sum form, we can also reformulate OT distance as an integral called L^p -Wasserstein distance for $p \in [1, \infty)$

$$W_p(r, c) = (\inf_{\gamma \in \Gamma(r, c)} \int_{M \times M} d(x, y)^p d\gamma(x, y))^{\frac{1}{p}} \quad (3)$$

Here, we can find that Eq. (2) is a special case of Eq. (3) with $p = 1$ [13].

4. Materials and method

In this section, we firstly introduce the subjects, image pre-processing method and brain network construction used in this work. Then, we introduce the proposed optimal transport based pyramid graph kernel and offer some necessary theoretical proofs for it. Finally, we introduce the optimal transport based pyramid graph kernel learning framework for brain disease classification, as shown in Fig. 2.

4.1. Subjects

Brain networks used in our experiment are based on rs-fMRI data from ABIDE database which has aggregated functional and structural brain imaging data collected from laboratories around the world to accelerate the understanding of the neural bases of autism. These rs-fMRI data were collected by five different imaging centers including NYU, Leuven, UCLA, UM and USM. There are respectively 164, 57, 74, 113, 60 subjects in NYU, Leuven, UCLA, UM and USM. All subjects have corresponding rs-fMRI and demographic information. The detailed demographic information is shown in Table 1. The detailed scan parameters, protocols, and procedures are shown on the ABIDE website¹. The rs-fMRI data in

¹ http://fcon_1000.projects.nitrc.org/indi/abide/.

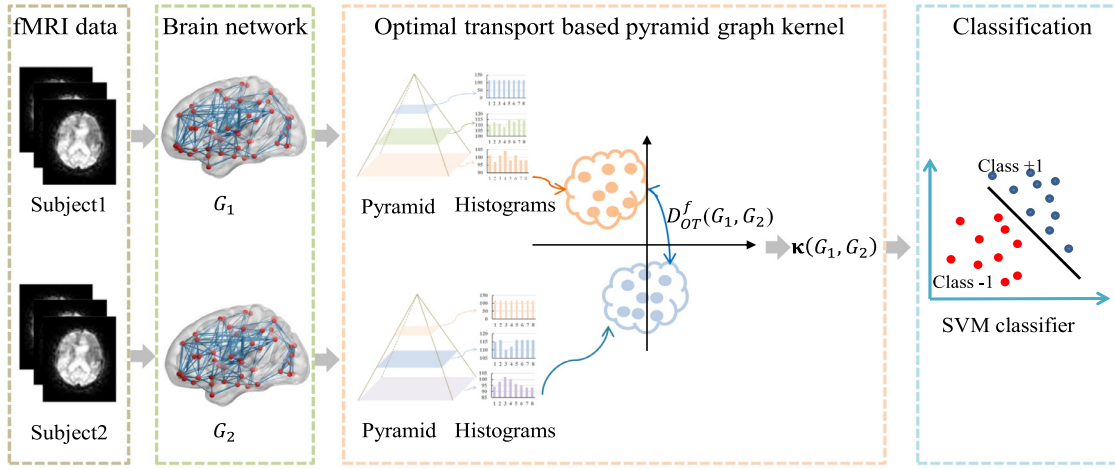


Fig. 2. Illustration of the optimal transport based pyramid graph kernel learning framework. f represents pyramid embedding. OT is optimal transport. SVM is support vector machine.

Table 1

Demographic information of the studied subjects from five imaging centers in the ABIDE database. The values are denoted as mean±standard deviation. M/F: male/Female. ASD is autism spectrum disorder. NC is normal control.

Center	ASD		NC	
	Age	M/F	Age	M/F
NYU	17.59±7.84	66/5	16.49±7.68	79/14
Leuven	13.10±4.79	21/4	18.80±9.00	24/8
UCLA	16.27±6.48	28/8	14.65±4.97	31/7
UM	17.05±8.36	43/5	17.35±7.12	56/9
USM	15.77±7.21	30/8	17.34±9.53	21/1

five imaging centers were obtained using different parameters and scanners, hence they should be regarded as five separate classification tasks.

4.2. Image processing and brain network construction

All rs-fMRI data are preprocessed by utilizing the Configurable Pipeline for the Analysis of Connectomes [27]. T1-weighted MR images of all subjects are segmented into white matter (WM), cerebrospinalfluid (CSF) and gray matter (GM). GM tissues of all subject are utilized to mask the corresponding fMRI to remove the possible effect of WM and CSF in time series of the fMRI. The first ten time points of each subject's rs-fMRI data were removed for signal equilibrium and ensuring subjects adapt to the scanning noise. The slice-timing corrections were performed by utilizing middle slice as reference frame. The head motion corrections were executed for adjusting the time series of images to correct each brain in the same position in every image. Nuisance signal regression and temporal filtering were performed. Each functional image was spatially normalized into standard Montreal Neurological Institute space by using a non-linear registration algorithm (i.e., advanced normalization tools, ANTS [28]).

The whole-brain subcortical and cortical structures were subdivided into 116 brain regions for all subjects based on Automated Anatomical Labeling (AAL) atlas [7]. Each brain region was treated as a region-of-interest (ROI), and mean rs-fMRI time series were computed via averaging blood oxygen level-dependent (BOLD) signals extracted from GM tissue among all voxels in each specific ROI. The Pearson correlation coefficient between mean time series of paired ROIs was then calculated for measuring the functional connectivity between paired ROIs. Accordingly, a 116×116 weighted brain functional network was created for each subject.

The related research revealed that the networks with average connection density interval of [25%,75%] could achieve higher classification performance [29]. In our work, we set the threshold value as 0.3, which corresponds to 30% network connectivity density, and then obtain the binary brain network for each subject.

4.3. Optimal transport based pyramid graph kernel

Euclidean distance is a popular distance. In this section, we use Euclidean distance as ground distance of OT and propose an optimal transport based pyramid graph kernel (OTPGK) which is a positive definite OT graph kernel in graph pyramid structures. The calculation of the proposed OTPGK mainly includes: (1) pyramid embedding, (2) optimal transport with entropic constraints, (3) OT distance on pyramids, and (4) optimal transport based pyramid graph kernel. Before introducing the contents related to OTPGK, we first present the definition and theorem related to positive definite OT graph kernel.

Definition 1 (OT graph kernel). Given a set of graphs $\mathcal{G} = \{G_1, G_2, \dots, G_N\}$ and OT distance D_{OT}^f between each pair of graphs is defined by Eq. (2). N is the number of graphs. f is graph embedding function which transform graphs into probability distributions or histograms. The OT graph kernel is defined as

$$\kappa_{OT}(G_i, G_j) = e^{-\lambda D_{OT}^f(G_i, G_j)} \quad (4)$$

where G_i and G_j are two arbitrary graphs, $G_i, G_j \in \mathcal{G}$, $\lambda > 0$. $D_{OT}^f(G_i, G_j)$ is the graph OT distance measuring the transport cost between graph G_i and G_j .

Obviously, OT graph kernel is based on Laplacian kernel. This kernel may not be positive definite. Because, OT distance D_{OT}^f is not isometric, the metric space it induces strongly depends on ground distance [30]. Here we offer the necessary and sufficient condition that OT graph kernel is positive definite as shown in Theorem 1.

Theorem 1. $\kappa_{OT}(G_i, G_j)$ is a positive definite kernel for all $\lambda > 0$ if and only if $D_{OT}^f(G_i, G_j)$ is conditionally negative definite (CND).

Proof. $D_{OT}^f(G_i, G_j)$ can be regarded as a metric function M defined on two graphs G_i and G_j . According to Theorem 5.2 in [31] and Theorem 2 in [32], the kernel $\exp(-\lambda M)$ is positive definite for all $\lambda > 0$ if and only if the metric function M is CND. Hence, $\kappa_{OT}(G_i, G_j)$ is positive definite if and only if $D_{OT}^f(G_i, G_j)$ is CND. $\square$

4.3.1. Pyramid embedding

Pyramid embedding, which transforms data (e.g., graph and image) into pyramid structures, is based on eigen-decomposition

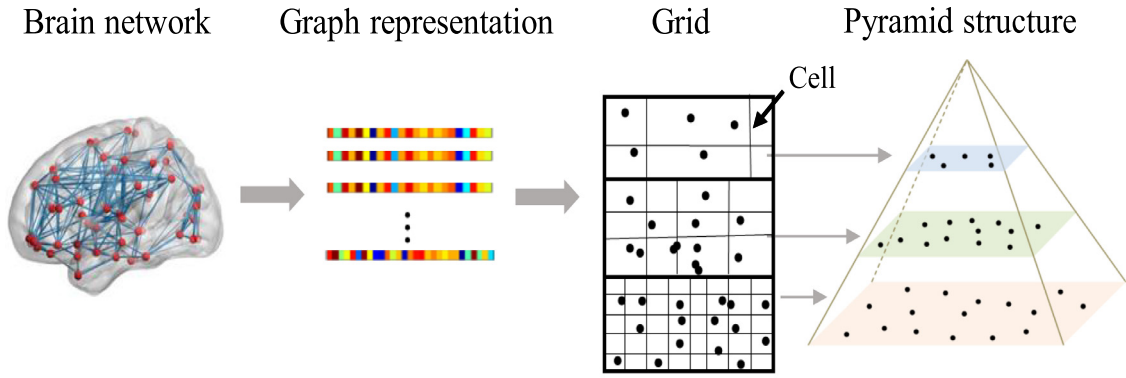


Fig. 3. Illustration of graph pyramid embedding scheme. The cells surrounded by thick black lines form a grid.

strategy. Each data is transformed into a bag-of-vectors using eigen-decomposition. Then, the vectors from bags-of-vectors are projected into multi-resolution histograms whose shape is like a pyramid structure. After obtaining the pyramid structures of data, the existing kernel methods based on pyramid structure used histogram intersections to measure the similarity of paired data. For example, in the research of matching node embedding for graph similarity, Nikolentzos et al. [9] used the histogram intersections of pyramid structure to design graph kernel. However, they ignored the geometric information at each layer of the pyramid. To address this problem, we present a graph pyramid embedding scheme to project the graphs into pyramid structure and use the OT distance to measure the similarity at each layer of the pyramid structures between pairwise graphs. Here, we will introduce a graph pyramid embedding scheme.

Graph pyramid embedding scheme. Each graph is embedded into d dimension vector space utilizing eigen-decomposition strategy for obtaining graph vector representation. In vector space, each node in graph corresponds to a point in d dimensional unit hypercube which can be regarded as a cubical space with d dimensions. We partition the points in d dimensional unit hypercube into multiple cells and put these cells on the grid. The grid is fitted with cells of increasing size to the d dimensional unit hypercube which has 2^l cells along each dimension. The d dimensional unit hypercube has d dimensions and each dimension has 2^l cells. Hence, the total number of cells in d dimensional unit hypercube is 2^{ld} . The illustration of graph pyramid embedding scheme is shown in Fig. 3. The steps of computing pyramid embedding are shown in Algorithm 1. In Fig. 3, the cells surrounded by thick black lines form a grid.

We obtain the pyramid structure of graph using pyramid embedding scheme, and then need to calculate the OT distance on pyramid structure. OT distance defined in Eq. (2) is the general OT distance, which is not isometric. The graph kernel based on this distance is not positive definite [13]. To acquire the positive definite graph kernel, we need to modify the general OT distance. Here, we add the entropic constraints to the OT distance.

4.3.2. Optimal transport with entropic constraints

We use OT with entropic constraints to calculate OT distance at each layer of the pyramid structure. The entropic constraints are added to $\Gamma(r, c)$ in Eq. (1), the new transportation plan set is defined as

$$\Gamma'(r, c) = \{P \in \Gamma(r, c) | KL(P || rc^T) = 0\} \quad (5)$$

where $KL(P || rc^T)$ is Kullback-Leibler divergence of $P, rc^T \in \Gamma(r, c)$, $KL(P || rc^T) = \sum_{ij} p_{ij} \log \frac{p_{ij}}{r_i c_j}$, p_{ij} is the element of P .

Theorem 2. For any matrix $P \in \Gamma(r, c)$, $\Gamma(r, c)$ contains all possible joint probabilities of (X, Y) , $p(X = i, Y = j) = p_{ij}$. Let $p(X =$

$$i) = r_i, p(Y = j) = c_j, \text{ then } KL(P || rc^T) = \sum_{ij} p_{ij} \log p_{ij} - \sum_i r_i \log r_i - \sum_j c_j \log c_j$$

Algorithm 1 Pyramid-embedding(G, L, d).

Input: graph G , the layers of pyramid L , vector space dimension d
Output: graph pyramid P

- 1: $U = \text{eigs}(G)$ % Eigen-decomposition of graph G , U is the eigenmatrix of graph G %
- 2: **for** $l = 1$ to L **do**
- 3: $i = 2^l$
- 4: $D = \text{zeros}(d, i)$ % The total number of cells is 2^{ld}
- 5: $T = \text{ceil}(U * i)$ % T is cell, its size along the dimension is doubled at each iteration %
- 6: $T(T == 0) = 1$ % The size of cell along the other dimensions are equal to 1 %
- 7: **for** $j = 1$ to $\text{size}(U, 1)$ **do** % $\text{size}(U, 1)$ is the row of eigenmatrix U %
- 8: **for** $k = 1$ to $\text{size}(U, 2)$ **do** % $\text{size}(U, 2)$ is the column of eigenmatrix U %
- 9: $D(k, T(j, k)) = D(k, T(j, k)) + 1$ % Fit a grid with cells of increasing size to the d -dimensional unit hypercube. %
- 10: $P(l) = D$ % Each layer of graph pyramid stores multi-resolution histogram %
- 11: **return** P

Proof. According to maximum entropy for multidimensional contingency tables, the independence table rc^T has entropy $E(rc^T) = E(r) + E(c)$, $E(rc^T) = -\sum_{ij} r_i c_j \log r_i c_j$, $E(r) = -\sum_i r_i \log r_i$, $E(c) = -\sum_j c_j \log c_j$. $\square$

$$-\sum_{ij} r_i c_j \log r_i c_j = -\sum_i r_i \log r_i - \sum_j c_j \log c_j$$

$$\begin{aligned} KL(P || rc^T) &= \sum_{ij} p_{ij} \log \frac{p_{ij}}{r_i c_j} \\ &= \sum_{ij} p_{ij} \log p_{ij} - \sum_{ij} r_i c_j \log r_i c_j \\ &= \sum_{ij} p_{ij} \log p_{ij} - \sum_i r_i \log r_i - \sum_j c_j \log c_j \end{aligned}$$

The optimal transport distance with entropic constraints can be defined as:

$$d'_{OT}(r, c) = \min_{P \in \Gamma'(r, c)} \langle P, M \rangle \quad (6)$$

Theorem 3. $d'_{OT}(r, c)$ is a distance metric and satisfies the axioms of distance metric.

Proof. Wasserstein distance W_p defined in Eq. (3) satisfies all three distance axioms [13]. $d'_{OT}(r, c)$ is a special case of W_p . Hence, $d'_{OT}(r, c)$ also satisfies the axioms of distance metric. $\square$

In this work, we use Euclidean distance as ground distance. The distance $d'_{OT}(r, c)$ based on Euclidean distance is CND.

Theorem 4. If ground distance d is Euclidean distance, d'_{OT} is CND.

Proof. When ground distance d is Euclidean distance, cost matrix M is Euclidean distance matrix, $M_{ij} = \|\psi(x) - \psi(y)\|_2^2$, $x, y \in \mathcal{X}$, $\mathcal{X}$ is a nonempty set. According to Lemma 5.5 in [31], M is a negative definite distance. $\square$

We set $E(P) = -\sum_{i,j} p_{ij} \log p_{ij}$, then $KL(P||rc^T) = E(r) + E(c) - E(P)$. When $KL(P||rc^T) = 0$, we have $E(P) = E(r) + E(c)$. Because $E(rc^T) = E(r) + E(c)$, we have $rc^T = P$. Subsequently, the optimal transport distance with entropic constraints is defined as $d'_{OT} = r^T M c$. M is negative definite, hence d'_{OT} is CND.

4.3.3. OT distance on pyramids

OT distance is a distance metric defined on probability distributions or histograms, which can capture the geometric information of the underlying metric space [33]. After using graph embedding scheme (e.g., pyramid embedding) to obtain graph representations regarded as histograms (i.e., probability distributions), we define the OT distance on these pyramids. Here we call this distance pyramid OT distance.

Definition 2 (Pyramid OT distance). Given arbitrary two graphs $G_i = (V_i, E_i)$, $G_j = (V_j, E_j)$, $i, j = 1, 2, \dots$, and graph embedding (i.e., pyramid embedding) scheme $f: G \rightarrow \mathbb{R}^{m \times n}$. The ground distance is $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Graph OT distance between G_i and G_j is defined as

$$D_{OT}^f(G_i, G_j) = d'_{OT}(f(G_i), f(G_j)) \quad (7)$$

4.3.4. Optimal transport based pyramid graph kernel (OTPGK)

Between a pair of graphs, we utilize Euclidean distance as ground distance and measure the similarity of histograms at each layer of the pyramid structure using the OT distance with entropic constraints. Then, we design the OT graph kernel at each layer of the pyramid structure. We calculate the summation of OT graph kernel from each layer of the pyramid structure and obtain the optimal transport based pyramid graph kernel (OTPGK).

Given two graph G_i and G_j , the graph OT distance at l layer of pyramid structure is calculated by

$$D_{OT}^{f_l}(G_i, G_j) = d'_{OT}(f_l(G_i), f_l(G_j)) \quad (8)$$

where $f_l(G_i)$ is pyramid embedding of graph G_i at l layer.

Definition 3 (Optimal transport based pyramid graph kernel).

For two graphs G_i and G_j , $D_{OT}^{f_l}(G_i, G_j)$ is the OT distance at l layer of pyramid structures between graph G_i and G_j . We assume that pyramid structure of each graph has L layers, $l \in L$. We define OT graph kernel κ_{OT}^l at l layer of pyramid structures and present optimal transport based pyramid graph kernel κ_{OTPGK} between graph G_i and G_j . The OT graph kernel at l layer of pyramid structures between graph G_i and G_j is defined as

$$\kappa_{OT}^l(G_i, G_j) = e^{-\lambda D_{OT}^{f_l}(G_i, G_j)} \quad (9)$$

where f_l is the pyramid embedding of graph at l layer.

Theorem 5. The graph kernel $\kappa_{OT}^l(G_i, G_j)$ is positive definite.

Proof. According to Theorem 4, we know that $D_{OT}^{f_l}(G_i, G_j)$ is CND. Based on Theorem 1, $\kappa_{OT}^l(G_i, G_j)$ is positive definite when $D_{OT}^{f_l}(G_i, G_j)$ is CND. $\square$

The optimal transport based pyramid graph kernel κ_{OTPGK} between graph G_i and G_j is defined as

$$\kappa_{OTPGK}(G_i, G_j) = \sum_{l=1}^L \kappa_{OT}^l(G_i, G_j) \quad (10)$$

Theorem 6. The optimal transport based pyramid graph kernel κ_{OTPGK} is positive definite.

Proof. $\kappa_{OT}^l(G_i, G_j)$ is positive definite. $\kappa_{OTPGK}(G_i, G_j)$ is the summation formula of $\kappa_{OT}^l(G_i, G_j)$, hence κ_{OTPGK} is positive definite.

The steps of computing optimal transport based pyramid graph kernel are summarized in Algorithm 2. $\square$

Algorithm 2 Compute optimal transport based pyramid graph kernel κ_{OTPGK} .

Input: Two graphs G_i, G_j , parameter λ , the layers of pyramid L , vector space dimension d

Output: OT graph kernel $\kappa_{OTPGK}(G_i, G_j)$

```

1:  $P_{G_i} = \text{Pyramid} - \text{embedding}(G_i, L, d)$ ; %Compute graph pyramid
   embedding of graph  $G_i$  and  $G_j$ 
2:  $P_{G_j} = \text{Pyramid} - \text{embedding}(G_j, L, d)$ ;
3:  $K = 0$ 
4: for  $l=1$  to  $L$  do
5:    $D_{OT}^{f_l}(G_i, G_j) = d'_{OT}(P_{G_i}^l, P_{G_j}^l)$ ; %Compute the OT distance at  $l$ 
     layer of pyramid%
6:    $\kappa_{OT}^l(G_i, G_j) = e^{-\lambda D_{OT}^{f_l}(G_i, G_j)}$  %Calculate OTPGK based on OT
     distance%
7:    $K = K + \kappa_{OT}^l(G_i, G_j)$ 
8:  $\kappa_{OTPGK} = K$ 
9: return  $\kappa_{OTPGK}$ 

```

4.4. Optimal transport based pyramid graph kernel learning framework

We utilize the image processing method introduced in materials and method to process the rs-fMRI data of all subjects and construct the brain functional network for each subject. In the brain functional network, nodes represent brain regions, and edges represent the functional connections between the paired brain regions. Given the brain functional networks of all subjects, we compute our proposed optimal transport based pyramid graph kernel on them with Algorithm 2 and apply the kernel matrix to support vector machine (SVM) classifier for ASD classification.

5. Results

In this section, we introduce the experimental settings and exhibit classification results from our proposed method and the competing methods using brain network data based on rs-fMRI.

5.1. Experiment settings

We perform binary classification tasks in five different ASD rs-fMRI datasets from different imaging centers including NYU, Leuven, UCLA, UM and USM. The experiments are performed in the thresholded brain functional networks of all subjects and the threshold value is 0.3. The layers of pyramid L is chosen from 1 to 4. The vector space dimension d is chosen from $\{2, 4, 6, 8\}$. We exploit SVM as the final classifier to perform the classification experiment. SVM is based on LIBSVM library. The tradeoff parameter C in the SVM is chosen from $\{10^{-5}, 10^{-4}, \dots, 10^5\}$. We perform

Table 2

Classification results of all methods. OTPGK is the proposed method. ACC: classification accuracy, BAC is balanced accuracy.

Method	Leuven			NYU			UCLA			UM			USM		
	ACC(%)	BAC(%)	AUC(%)	ACC(%)	BAC(%)	AUC(%)	ACC(%)	BAC(%)	AUC(%)	ACC(%)	BAC(%)	AUC(%)	ACC(%)	BAC(%)	AUC(%)
SP [18]	57.9	52.0	54.2	56.7	50.0	60.2	61.3	59.4	58.3	57.5	50.0	56.2	63.3	50.0	61.7
RW [19]	57.9	55.1	53.3	75.0	72.3	74.5	58.1	58.1	56.4	76.1	72.4	73.2	75.0	66.9	71.3
WL-SP [23]	47.4	45.3	44.2	44.5	42.6	45.1	55.4	55.4	52.8	55.8	54.2	52.3	50.0	46.2	48.5
WL-ST [23]	77.2	74.0	73.3	78.3	76.2	77.4	74.3	74.4	72.6	78.7	75.6	76.2	71.7	61.4	68.7
GH [35]	75.4	72.0	71.3	57.9	51.4	54.2	73.2	70.4	71.6	61.9	55.2	56.3	63.3	50.0	54.4
PM [9]	87.7	86.0	85.1	80.2	76.2	78.9	83.7	83.3	81.2	77.2	75.4	73.7	75.7	71.3	72.8
SKL [36]	77.2	74.0	73.2	72.6	68.3	69.1	74.3	73.6	70.7	72.6	67.7	70.9	76.7	68.2	68.7
WWL [13]	81.4	79.4	78.5	73.6	70.3	71.1	77.7	76.2	72.0	71.5	69.4	67.4	72.1	70.7	70.3
Tree+ [34]	56.1	50.0	54.2	56.7	50.0	55.6	39.8	38.2	40.1	57.5	50.0	51.2	63.3	50.0	60.7
DIFFPOOL [37]	83.3	80.1	79.4	72.4	71.6	70.3	80.5	79.1	79.8	74.4	73.2	73.1	73.8	71.3	72.6
OTPGK	92.9	92.0	90.3	81.1	78.2	79.1	86.5	86.1	85.4	79.7	76.0	77.2	78.3	74.5	75.6

leave-one-out cross-validation for all classification tasks. We evaluate the performances of all methods via calculating classification accuracy (ACC), balanced accuracy (BAC, i.e., average classification accuracy on either class), and area under receiver operating characteristic curve (AUC).

We compare the proposed optimal transport based pyramid graph kernel with the state-of-the-art methods involving shortest path kernel (SP) [18], Weisfeiler-Lehman shortest path kernel (WL-SP) [23], Weisfeiler-Lehman subtree kernel (WL-ST) [23], pyramid match kernel (PM) [9], random walk kernel (RW) [19], Wasserstein Weisfeiler-Lehman graph kernel (WWL) [13], truncated tree based graph kernels (Tree++) [34], GraphHopper kernel (GH) [35], and sub-network kernels (SKL) [36], DIFFPOOL [37]. WL-ST and WL-SP belong to the Weisfeiler-Lehman graph kernel framework which is based on the Weisfeiler-Lehman scheme proposed by Shervashidze et al. [23]. WWL is proposed by Togninalli et al [13]. They combine the Wasserstein distance with the Weisfeiler-Lehman scheme to measure the similarity between a pair of graphs. PM measures the graph similarity using the intersections of multi-resolution histograms which look like a pyramid. SP, GH and Tree++ are based on path kernel. SP measures the similarity of pairwise graphs by computing the matched shortest path. GH measures pairwise graph similarity by counting sub-path similarities. Tree++ measures the graph similarity using super path containing truncated breadth-first search trees rooted at the vertices in a path. RW counts the number of matching random walks for measuring the similarity between paired graphs. SKL measures the similarity of paired networks by calculating the similarity between sub-networks from nodes. DIFFPOOL is graph neural network with differentiable pooling on hierarchical graph representation.

5.2. Brain disease classification

We compare the proposed optimal transport based pyramid graph kernel with state-of-the-art graph kernels in five ASD brain network data constructed by five different rs-fMRI datasets from different imaging centers including NYU, Leuven, UCLA, UM and USM. The classification results of all methods are summarized in Table 2. As shown in Table 2, the proposed graph kernel outperforms the competing methods in five classification tasks. Specifically, the proposed method yields the accuracies of 92.9%, 81.1%, 86.5%, 79.7%, and 78.3% for ASD vs. NC classification in Leuven, NYU, UCLA, UM and USM, while the best accuracies of the control methods are 87.7%, 80.2%, 83.7%, 78.7%, and 76.7%, respectively. Also, the BAC of our method is respectively 92.0%, 78.2%, 86.1%, 76.0%, and 74.5% in five classification tasks, while the best BAC of competing methods is respectively 86.0%, 76.2%, 83.3%, 75.6%, and 71.3%. In addition, our method yields the AUC of 90.3%, 79.1%, 85.4%, 77.2%, and 75.6% for ASD vs. NC classification in Leuven, NYU, UCLA, UM and USM, while the best AUC of the control meth-

ods are 85.1%, 78.9%, 81.2%, 76.2%, and 72.8%, respectively. These classification results demonstrate that our proposed optimal transport based pyramid graph kernel does well in distinguishing ASD from NC compared with state-of-the-art graph kernels.

Network threshold values will affect brain network structures. To verify the classification performance of our proposed graph kernel, we conduct the classification experiments on five ASD brain network data with different threshold values (i.e., $T=\{0.3, 0.35, 0.4, 0.45, 0.5\}$). The corresponding average brain connectivity density of each thresholded brain network is situated in the interval [30%,70%] [36]. The related research revealed that the networks with average connection density interval of [25%,75%] could achieve higher classification performance [36]. We compare the proposed graph kernel with the competing methods using the different thresholded brain network data, the classification results are shown in Fig. 4. From Fig. 4, we can find that our proposed graph kernel outperforms the competing methods in brain networks with different threshold values.

The results from Table 2 and Fig. 4 show that the proposed OTPGK is better than the competing methods including GNN in the classification tasks of ASD brain functional networks. We infer that the classification performances of the proposed method are related to pyramid structure which assigns nodes in brain functional network to the different hierarchies of hierarchical architecture by performing graph pyramid embedding scheme, as shown in Fig. 3. At same hierarchical architecture, nodes are considered to have the same importance in local structure characteristics. The node importance can be helpful for the proposed OTPGK to obtain excellent performance in ASD classification tasks.

5.3. Important brain regions

Before computing pyramid embedding, we use graph eigen-decomposition to transform each brain network into a vector matrix which is the vector representation of the brain network. Each row of vector matrix corresponds to the brain region's embedding, which is also a vector. We use exponential kernel defined in Eq. (9) to measure the similarity of brain regions between a pair of brain networks. The distance $D_{OT}^{f_i}$ is replaced by Euclidean distance between nodes. We calculate the kernel for each brain region, and then the kernels for all brain regions are applied to SVM classifier for brain disease classification in five datasets (i.e., Leuven, NYU, UCLA, UM, and USM). We obtain the classification accuracy of each ROI (i.e., brain region) for ASD classification. Finally, we rank these ROIs according to their classification accuracy and then select top 10 ROIs with the highest classification accuracy. The selected ROIs include left thalamus (THA.L), right thalamus (THA.R), left hippocampus (HIP.L), right hippocampus (HIP.R), right fusiform gyrus (FFG.R), left superior temporal gyrus (STG.L), right anterior cingulate-paracingulate gyri (ACG.R), and left amygdala (AMYG.L),

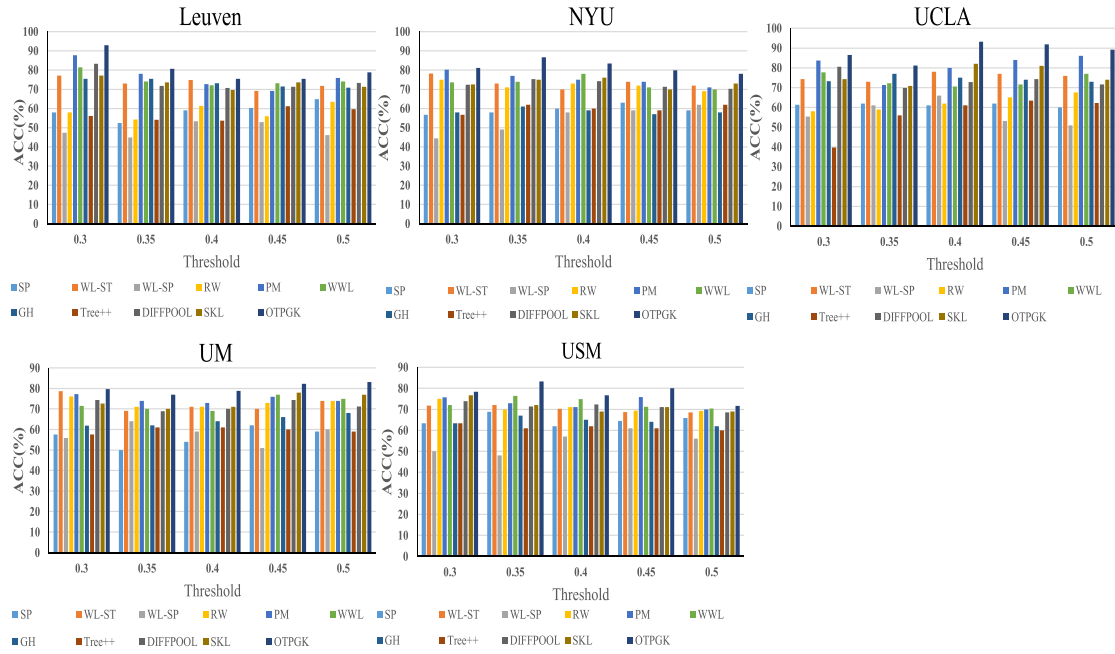


Fig. 4. Classification results in brain networks with different thresholds (i.e., $T = \{0.3, 0.35, 0.4, 0.45, 0.5\}$). ACC: classification accuracy.

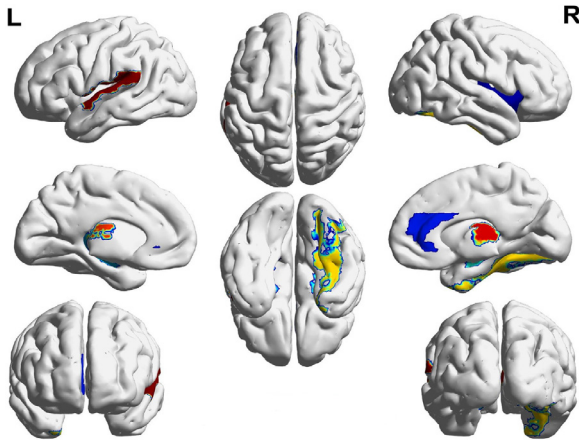


Fig. 5. The important ROIs are selected by our proposed graph kernel. The color represents brain regions.

Table 3

Full name and abbreviation of important brain regions .

Index	ROI	Full name
1	THA.L	Left thalamus
2	THA.R	Right thalamus
3	HIP.L	Left hippocampus
4	HIP.R	Right hippocampus
5	FFG.R	Right fusiform gyrus
6	STG.L	Left superior temporal gyrus
7	ACG.R	Right anterior cingulate and paracingulate gyri
8	AMYG.L	Left amygdala
9	INS.L	Left insula
10	INS.R	Right insula

left insula (INS.L), and right insula (INS.R). Fig. 5 exhibits all these ROIs. The full name and abbreviation of these important brain regions are shown in Table 3. These ROIs are consistent with those reported in the previous studies related to ASD [4].

5.4. Statistical differences of pyramid structures

In order to investigate the statistical differences of pyramid structures in brain networks between normal controls and ASD patients, we select 16 ASD patients and 14 normal controls from the UCLA dataset, and then apply the proposed optimal transport based pyramid graph kernel to kernel based statistical framework [38]. There are 8 male subjects and 8 female subjects in ASD patients, and 7 male subjects and 7 female subjects in normal controls. There are no statistical differences in age and gender between 16 ASD patients and 14 normal controls. The kernel based statistical framework is utilized to investigate the statistical differences of topological structures in brain networks. The detailed introductions can refer to the literature [38].

We construct the different pyramid structures of brain functional networks for ASD patients and normal controls. These pyramid structures have different layers including two layers ($L = 2$), three layers ($L = 3$), and four layers ($L = 4$), as shown in Fig. 6. From Fig. 6, we can find that there are significant differences in pyramid structures between ASD patients and normal controls. The p value is respectively 0.0131, 0.0124, and 0.0121 under significant level $\alpha = 0.05$.

Using graph pyramid embedding on brain networks, the nodes (i.e., brain regions) of brain networks are assigned to the different layers of pyramid structure. We calculate the number of the matched nodes at each layer of pyramid structure between ASD patients and normal controls. We find that the number of the matched nodes at the bottom of the pyramid structure is larger than those in the middle of the pyramid structure and those at the top of pyramid structure. Briefly, the numbers of the matched nodes decrease gradually from the bottom to the top of pyramid structure as shown in Fig. 7. Compared to normal controls, the unmatched nodes in different hierarchies of pyramid structures of ASD patients include right thalamus (THA.R), left hippocampus (HIP.L), right hippocampus (HIP.R), right fusiform gyrus (FFG.R), left superior temporal gyrus (STG.L), right anterior cingulate-paracingulate gyri (ACG.R), and left amygdala (AMYG.L), left insula (INS.L), and right insula (INS.R). These nodes are con-

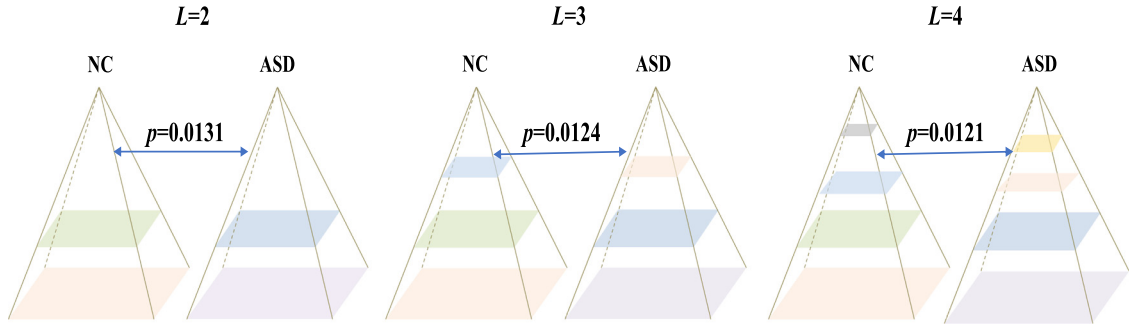


Fig. 6. Statistical differences in pyramid structures of brain networks. Significant level $\alpha = 0.05$. NC is normal control. ASD is autism spectrum disorder. L is the layer number of pyramid structure.

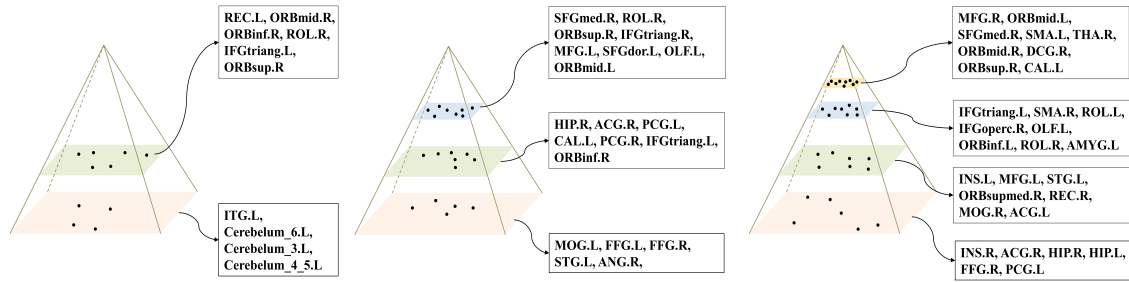


Fig. 7. Unmatched nodes in different hierarchies of pyramid structures.

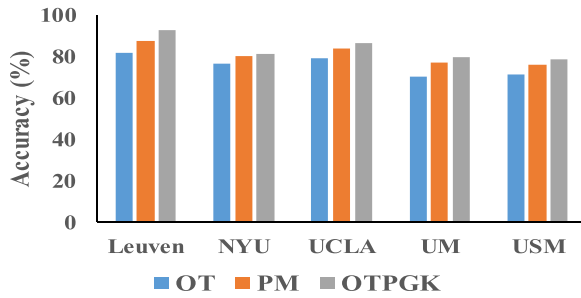


Fig. 8. Ablation study in Leuven, NYU, UCLA, UM and USM. OT is OT graph kernel. PM is PM graph kernel. OTPGK is the proposed optimal transport based pyramid graph kernel.

sistent with the important brain regions in Fig. 5. Meantime, we can also find that the nodes (e.g., HIP.L and HIP.R) related to brain diseases (i.e., ASD) exist in the pyramid structure with more hierarchies. We infer that these nodes affect the pyramid structures of brain networks and result in the statistical differences of pyramid structures between ASD patients and normal controls.

5.5. Ablation study

In this section, we conduct an ablation study to evaluate the usefulness of OT distance and pyramid structure on the performance of the proposed method for ASD vs. NC classification in Leuven, NYU, UCLA, UM and USM. To assess the contribution of combining OT distance and pyramid structure, we perform classification tasks using OT graph kernel and pyramid graph kernel. OT graph kernel is constructed by directly applying OT distance to brain networks. The pyramid graph kernel is PM kernel. We apply these two graph kernels to Leuven, NYU, UCLA, UM and USM for ASD vs. NC classifications. The classification results are shown in Fig. 8.

From Fig. 8, we can see that OTPGK kernel is better than OT kernel and PM kernel in three classification tasks. The classification

results from PM kernel are better than those of OT kernel. These results indicate that combining OT distance and pyramid structure can help improve the classification performance in ASD vs. NC classification tasks.

6. Discussion

6.1. Significance of results

The brain network has attracted extensive attention in studies related to brain disease. In brain network analysis, measuring the similarity of paired brain networks is a challenging task. Graph kernel offers an effective tool for processing this challenge. However, most of the existing graph kernels are built on single layer graph and neglect the hierarchical architecture information of brain network. In this paper, we have constructed a graph kernel based on hierarchical architecture information of brain networks and applied it to the classification tasks of brain disease (i.e., ASD). The experimental results on five real ASD datasets demonstrate that, compared with the state-of-the-art methods, our proposed graph kernel can significantly improve the classification performances.

In addition, some important ROIs have been identified by using exponential kernel (i.e., Eq. (9)) on brain regions. These ROIs involve thalamus, hippocampus, fusiform gyrus, superior temporal gyrus, anterior cingulate-paracingulate gyri, amygdala, and insula which have been reported in the previous studies related to ASD. For example, in the hippocampus and amygdala, children with ASD have an abnormal development pattern compared with normal children [39]. Functional activity in the insula has been reported to be abnormal during social tasks in ASD [40].

The related studies showed that the functional connectivities between identified brain regions in ASD group are significantly different from those in NC group. For example, Durkut et al. [40] have found that ASD patients have an alteration in the functional connectivity of insula compared with NC subjects. Xu et al. [39] found that functional connectivities between amygdala and hippocampus were different from those of NC.

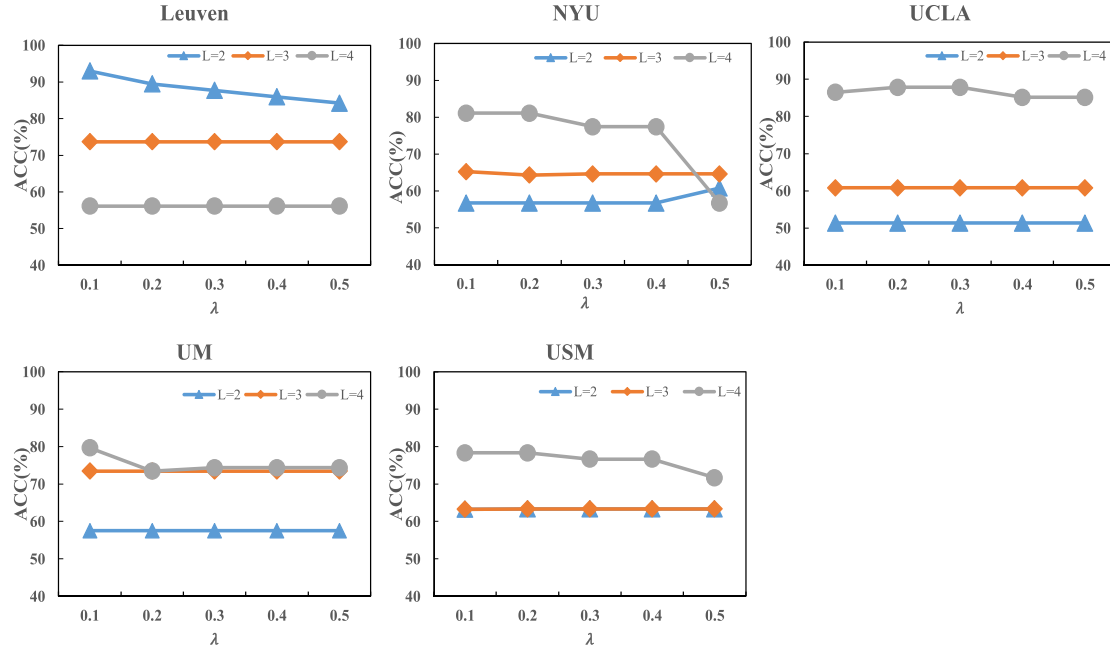


Fig. 9. The classification performance of our proposed graph kernel using different parameters (i.e., L and λ). ACC: classification accuracy.

These changes in the functional connectivities suggest that the edge connection information between some brain regions in the brain network may be disrupted by brain disease (e.g., ASD). These results may also imply that these disrupted functional connectivities may result in the changed hierarchical architecture information of brain network. Our experimental results demonstrate that our proposed optimal transport based pyramid graph kernel can effectively classify ASD from NC by making full use of hierarchical architecture information in brain functional network, which can also provide the empirical evidence for disorder network connectivity pattern in ASD.

6.2. Effect of parameters

In our proposed optimal transport based pyramid graph kernel, there exist two parameters including L and λ as shown in Eq (9) and (10). λ is the hyper-parameter of kernel function defined in Eq (9) and affects the value of kernel function. L is the layer number of graph pyramid structure, and controls the scale of pyramid structure. In order to investigate the influence of these two parameters on classification performance of our method, we test the different values for λ and L . The layer L is from $\{2, 3, 4\}$. The hyper-parameter λ is from 0.1 to 0.5. Fig. 9 exhibits the classification results with respect to the parameter L and λ on five datasets (i.e., Leuven, NYU, UCLA, UM, and USM). In Fig. 9, each curve represents the classification accuracy of the proposed graph kernel with different values of λ , when L is given.

From Fig. 9, we can find that most of the curves varied with the value of parameter λ are very smooth, when parameter L is fixed. These results indicate that the proposed method is robust to the parameter λ . Moreover, we find that, with a fixed λ , the classification performances are largely influenced by the different values of L . These classification results show that the selection of L (i.e., the layer number of graph pyramid structure) is very important for our proposed optimal transport based pyramid graph kernel. This is reasonable since the layer number L controls the scale of pyramid structure in each brain network, and thus influences the similarity measurement between paired brain networks.

6.3. Importance of network topology

In our work, the graph pyramid structure is a kind of hierarchical architecture which belongs to graph topological structure, and is used to calculate graph kernel for measuring brain network similarity. To estimate the importance of the topological structure of brain network, we conduct another experiment to examine the classification performance by directly utilizing adjacency matrices of brain functional networks. Specifically, we transform the adjacency matrices of brain functional networks constructed from rs-fMRI data of all subjects into vectors. We utilize t -test for feature selection in training set. Then, we apply the selected features from testing set to the SVM classifier. In five tasks of ASD vs. NC classification, this method yields accuracies of 50.8%, 65.0%, 51.0%, 58.0%, and 55.0%, respectively. These results are not better than most of the methods based on graph kernels, suggesting that network topological structure can help improve the classification performance of brain diseases.

7. Conclusion

In this paper, we propose a new graph kernel called optimal transport based pyramid graph kernel for measuring brain network similarity. Different from the existing methods for brain network analysis, our proposed graph kernel can make full use of the hierarchical architecture information of brain network to diagnose brain diseases (e.g., ASD). The proposed pyramid structure can be regarded as a new representation which is used to detect the changed network structure of patients with brain diseases. The experimental results in the classification tasks of brain diseases show that the proposed graph kernel outperforms the state-of-the-art graph methods. All these evidences demonstrate the promise of our proposed method for clinical studies and graph kernels. However, our work still has a limit. In the results section, we utilize the specific threshold values (i.e., $T=\{0.3, 0.35, 0.4, 0.45, 0.5\}$) to sparse brain networks and do not use other network construction models to construct brain networks. Different brain network structures constructed with different network models can provide different perspectives for describing brain networks. Hence, the net-

work structures from different network models should have their importance for measuring the similarity between paired brain networks, which are not considered in current research. In the future study, we will validate the effectiveness of our proposed graph kernel in different network structures.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

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